

Linear Maps, Ridge Regression and SVD

A compact study guide for understanding encoder-to-Hub mappings

Encoder input: X | Hub target: H | Learned map: W | Prediction: $\hat{H} = XW$

What this guide connects

Concept	Core purpose	In your Hub experiment
Linear regression	Find a linear map that minimizes error	Map an encoder's vectors into Hub coordinates
Ridge regression	Stabilize that map with an L2 penalty	Control correlated or weak encoder directions
SVD	Reveal ranked directions of variation	Build and compress the concatenated multi-encoder Hub
Whitening	Equalize retained component variance	Prevent high-variance Hub axes from dominating similarity

Central idea

A vector is one point. A representation space is the coordinate system and geometry containing possible vectors. A linear map changes coordinates between spaces; SVD discovers useful axes; ridge makes the learned map more stable.

1. Notation and spaces

Symbol	Meaning	Example
in	belongs to / has this shape	$X \text{ in } \mathbb{R}^{(8533 \times 768)}$
$\mathbb{R}^d$	all real-valued vectors with d coordinates	$x \text{ in } \mathbb{R}^{768}$
X^T	transpose: rows and columns are swapped	8533×768 becomes 768×8533
I	identity matrix: ones on the diagonal	I has size $d \times d$
A^{-1}	matrix inverse: undoes A	$A^{-1}A = I$

Vector versus space

$x \text{ in } \mathbb{R}^{768}$ one encoder vector $X \text{ in } \mathbb{R}^{(8533 \times 768)}$ 8,533 vectors stored as rows

Two encoders may both output 768-dimensional vectors yet still have different practical representation spaces. Their vectors can have different distances, angles, clusters, anisotropy and neighborhoods.

Shape check for an encoder-to-Hub map

$X: 8533 \times 768$ $W: 768 \times 512$ $\hat{H} = XW: 8533 \times 512$

The inner dimensions must match: 768 meets 768. The remaining dimensions determine the result: 8,533 samples by 512 Hub coordinates.

2. Linear regression and ridge regression

Both methods learn the same kind of linear prediction: $\hat{H} = XW$. The difference is the objective used to choose W .

Method	Objective	Effect
Ordinary linear regression	$\min \ H - XW\ ^2$	Prioritizes training fit
Ridge regression	$\min \ H - XW\ ^2 + \lambda \ W\ ^2$	Balances fit with smaller coefficients

What lambda controls

lambda = 0 ordinary least squares small lambda mild shrinkage large lambda stronger shrinkage and greater stability

Lambda is not learned automatically in the closed-form equation. It is normally selected using held-out validation or cross-validation. Standardizing features makes lambda comparisons meaningful.

Closed-form ridge man

$$W = (X^T X + \lambda I)^{-1} X^T H$$

Part	Interpretation
$X^T H$	Raw relationships between encoder dimensions and Hub dimensions
$X^T X$	Input scale, correlation and redundancy structure
λI	Adds lambda to every diagonal entry; stabilizes weak directions
inverse	Corrects for the input geometry and isolates W

3. Why the inverse matters

Start with the ridge normal equation:

$$(X^T X + \lambda I)W = X^T H$$

Let $A = X^T X + \lambda I$. Then $AW = X^T H$. To isolate W , multiply both sides by A^{-1} :

$$A^{-1}AW = A^{-1}X^T H \quad IW = A^{-1}X^T H \quad W = A^{-1}X^T H$$

For numbers, division isolates the unknown: $5w = 20$ gives $w = 5^{-1}20 = 4$. Matrix division is represented by the inverse. The inverse does more than rescale: it untangles correlated input directions.

Why $X^T H$ alone is not the map

$X^T H$ measures raw association. A high-variance input dimension can appear more influential simply because it is larger. Correlated dimensions can repeat the same information. Multiplication by the inverse corrects these effects so W represents conditional, not merely raw, relationships.

Practical implementation

Numerical software generally solves the linear system directly rather than explicitly forming the inverse. Conceptually the inverse explains the solution; computationally a solver is usually more stable.

Ridge and invertibility

If X has redundant dimensions, $X^T X$ may be singular or nearly singular. Adding λI lifts its eigenvalues away from zero, making the system better conditioned and reducing coefficient explosions.

4. SVD: discovering the natural axes

$$C = U \Sigma V^T$$

Component	Meaning
V^T	Directions in the original feature space; each is a combination of original dimensions
Σ	Singular values, ordered from largest to smallest; strength of each direction
U	Sample coordinates in normalized singular directions
$U \Sigma = CV$	Unwhitened coordinates of samples on the discovered axes

Importance

In SVD, importance means variation captured, not guaranteed semantic usefulness. The variance associated with direction i is proportional to σ_i^2 .

$$\text{explained_variation}_i = \sigma_i^2 / \sum_j (\sigma_j^2)$$

Example: if $\sigma_1 = 10$ and $\sigma_2 = 1$, their squared strengths are 100 and 1. Direction 1 therefore accounts for about 99% of the variation.

Important caution

High variance can reflect semantics, but it can also reflect scale, modality differences, anisotropy, dataset bias or noise. Retaining a component is a statistical decision; held-out evaluation determines whether it is useful.

5. SVD in the shared Hub

First, independently standardize each encoder space. Then concatenate their aligned representations column-wise:

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C = [X_img_small | X_img_base | X_img_large | X_bge | X_gpt2 | X_bert | X_sberr]
```

Apply SVD and retain the leading $k = 512$ components:

```
C = U Sigma V^T Z = U_512 Sigma_512 = C V_512 unwhitened PCA coordinates
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Whitening divides each retained coordinate by its singular-value scale:

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H = Z Sigma_512^(-1) approximately U_512, depending on normalization
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Stage	Purpose
Standardize each space	Reduce domination caused only by encoder scale
Concatenate aligned rows	Let SVD observe all seven encoders jointly
Keep leading 512	Retain strong directions and discard much low-variation structure
Whiten	Give retained Hub axes equal variance
Fit W_k independently	Allow each encoder to enter the frozen shared Hub

After the Hub is frozen, each encoder k receives its own ridge map:

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X_k W_k ≈ H
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The Hub is shared; the entry maps are encoder-specific. W_k looks only at encoder k 's vectors and the already-built H .

6. Near-null singular values

A near-null singular value is extremely close to zero:

$\sigma_i \approx 0$

It indicates almost no independent variation in that direction. This often happens when dimensions are nearly redundant, for example $x_2 \approx x_1$.

Why it is dangerous

$\sigma_i = 0.000001$ implies $1/\sigma_i = 1,000,000$

Inverse operations or whitening can therefore amplify a microscopic amount of noise into a large effect. An exact zero signals exact linear dependence; a near-zero value signals approximate dependence and poor numerical conditioning.

How ridge protects the map

without ridge: behavior involves $1/\sigma_i^2$ with ridge: direction factor = $\sigma_i / (\sigma_i^2 + \lambda)$

Lambda prevents a weak singular direction from producing an enormous coefficient. For Hub whitening, the corresponding protection is to retain only reliable leading components, impose a threshold, or use regularized whitening.

Final mental model

Question	Answer
What is a space?	A coordinate system and geometry in which vectors live
What does SVD do?	Finds and ranks the data's natural variation directions
What is a singular value?	The strength of variation along one discovered direction
What is near-null?	A direction with almost no reliable independent information
Why ridge?	To prevent weak or correlated directions from destabilizing W
Why the inverse?	To correct input geometry and solve for the actual map